

AlgoLabs Assignment - 1

Finance Dashboard: Stock Market Monitoring Platform

Project Report

1. Project Architecture

Finance Dashboard is a full-stack application that collects, stores, and visualizes Indian stock market data.

- **Data:** yFinance is used to fetch stock prices and financial metrics.
- **Backend:** FastAPI provides REST APIs for accessing stock information.
- **Database:** SQLite stores company and market data.
- **Frontend:** Streamlit with Plotly provides an interactive dashboard.

Flow:

yFinance → Database → FastAPI → Streamlit Dashboard

2. APIs Used

External API:

- yFinance for stock prices, fundamentals, and historical data.

REST APIs:

- /stocks – Fetch all stocks
 - /stocks/{ticker} – Fetch specific stock details
 - /market-summary – View market overview
 - /history/{ticker} – Fetch historical prices
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3. Database Design

SQLite stores:

- Stock ticker and company name
- Price
- Market capitalization
- P/E ratio
- EPS
- Trading volume

- OHLC price data
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4. Features Implemented

- Tracks 20+ Indian companies
 - Interactive stock dashboard
 - Historical price charts
 - Company comparison
 - Fundamental analysis
 - Search and sorting
 - CSV export functionality
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5. Challenges Faced

- Handling incomplete market data from APIs
 - Designing an efficient database structure
 - Connecting backend APIs with the dashboard
 - Creating meaningful financial visualizations
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6. Future Improvements

- User watchlists and portfolio tracking
 - Stock screening tools
 - Technical indicators
 - AI-based insights
 - Automated alerts
 - Cloud database migration
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Summary:

Finance Dashboard provides a simple platform to monitor Indian stock market performance by combining data collection, backend APIs, database storage, and interactive visualizations.